# Important Notes
# 重要说明与打印建议

1. All geometry was adjusted to standard slicer settings and basic PLA, you can add walls and infill for stronger parts, using PETG will also enforce parts.

2. In most cases flat bolts can be swapped to countersunk bolts.

3. We used TPU by bambulab for tires, they should print in 7H, standard settings for TPU in bambulab slicer are recommended.

4. Main grey color used in videos is light grey 10104 PLA by bambu.

5. We highly recommend using 2F modules after base and before roof, when you are using modifications.

6. Holes for casters are M8, make sure to check it before purchasing them.

提示：1. 所有结构针对标准切片与基础 PLA 优化，增加壁厚与填充可强化受力；2. 轮胎推荐使用 TPU 耗材；3. 底座之后及顶盖之前，推荐使用 2F 模块过渡。

# VAULT



3D Printed Modular Storage System

*Instructions Manual*

# 五金配件清单 (Hardware List)



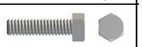
## DRAWERS

| | | | | |
|---|---|---|---|---|
| 1. | M4 14MM bolts | x4 | *per floor* |  |
| 2. | M6 50MM bolts | x2 | *per roof handle* |  |

## SUSTAINERS

| | | | | |
|---|---|---|---|---|
| 1. | M4 10MM bolts | x8 | *per module* |  |
| 2. | M4 14MM bolts | x16 | *per module* |  |
| 3. | M4 20MM bolts | x7 | *per module* |  |

## MODIFICATIONS

| | | | | |
|---|---|---|---|---|
| 1. | M6 30MM bolts | x4 | *per pull-out handle* |  |
| 2. | M6 35MM bolts | x2 | *per pull-out handle* |  |
| 3. | M6x8x8 heat insert | x2 | *per pull-out handle* |  |
| 4. | M4 60MM bolts | x8 | *per static handle* |  |
| 5. | M4 40MM bolts | x4 | *per pegboard board* |  |
| 6. | M4 60-100MM bolts | x4 | *per backbone board* |  |
| 7. | M16 504MM rod | x1 | *per XL base* |  |
| 8. | M16 414MM rod | x1 | *per L base* |  |
| 9. | M16 338MM rod | x1 | *per M base* |  |
| 10. | M16 248MM rod | x1 | *per S base* |  |
| 11. | M16 N-MM rod | x2 | *per pull-out handle* |  |
| 12. | M4 25MM bolts | x6 | *per wheel* |  |
| 13. | M6 90MM bolts | x2 | *per foot* |  |
| 14. | 6205ZZ bearing | x1 | *per wheel* |  |
| 15. | SCSAW stopper | x1 | *per wheel* |  |

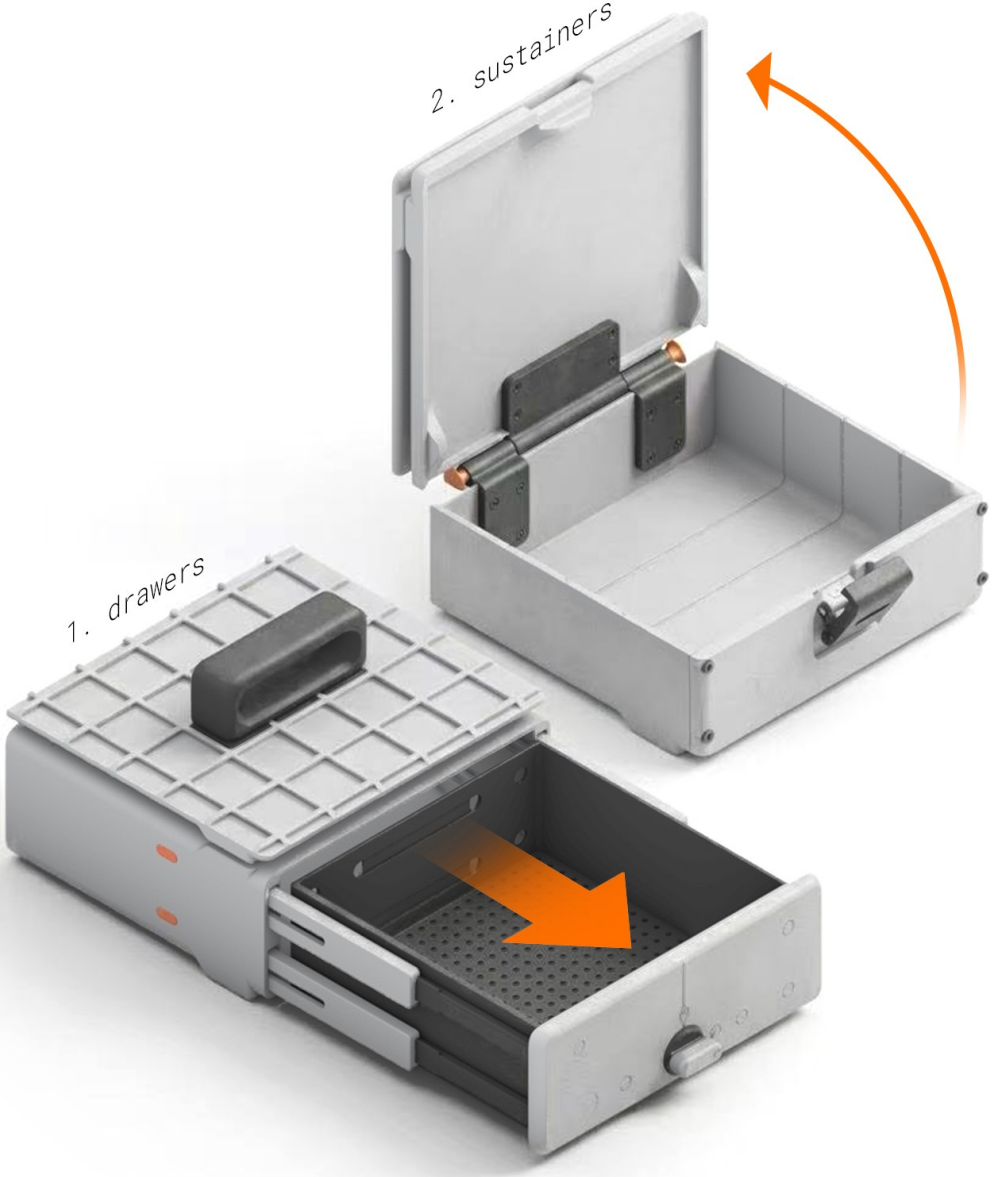
# VAULT 系统的五种使用形态





# VAULT 两大产品类型：抽屉款与收纳盒款



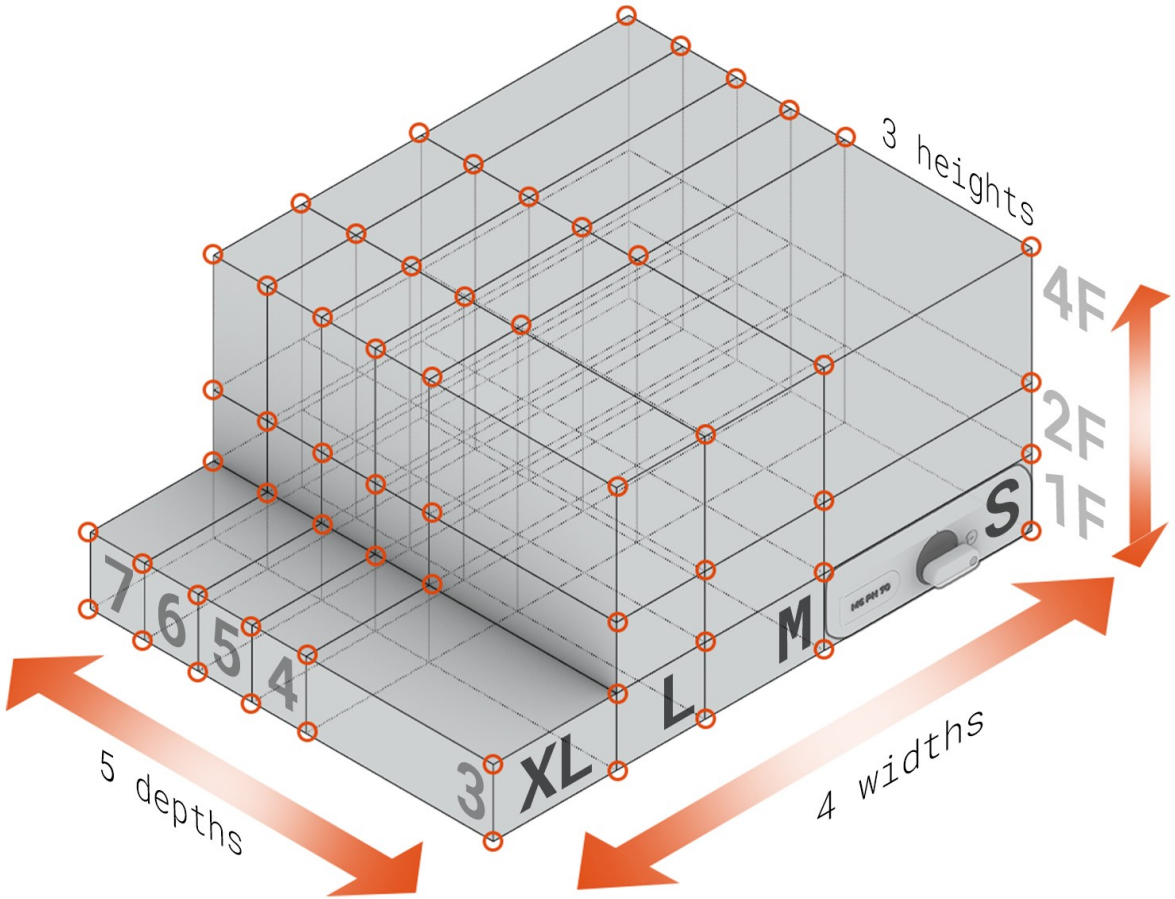



* 模型归档文件分为抽屉款与收纳盒款两大部分，请根据打印需求选用

*

# S / M / L / XL 规格 — 50 种尺寸衍生

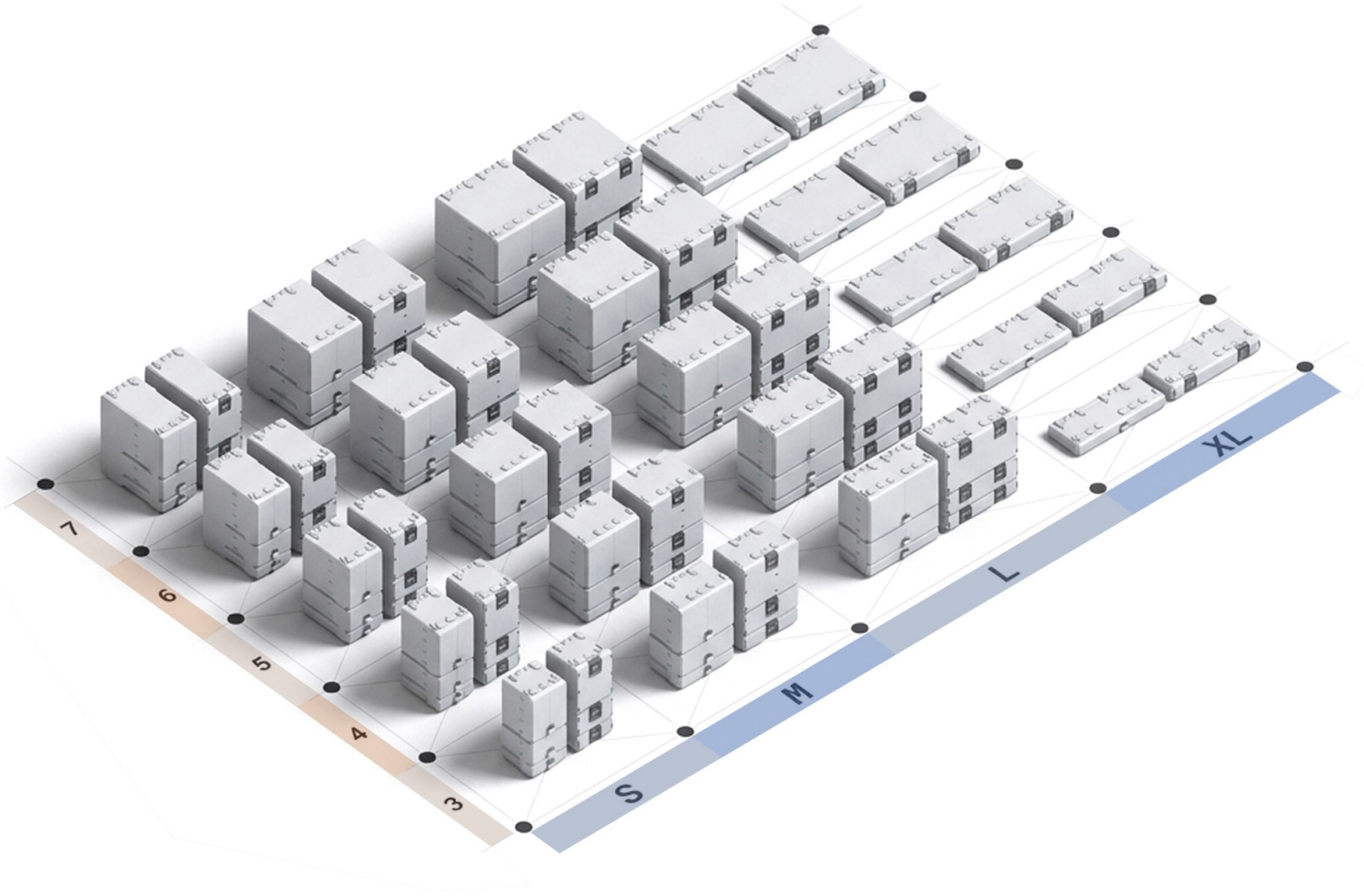





* 深度由 Gridfinity 单元（42mm）决定；宽度由 Gridfinity 单元与打印尺寸决定；高度包含 1F/2F/4F（无 3F 避免间隙，均被 2 整除）。

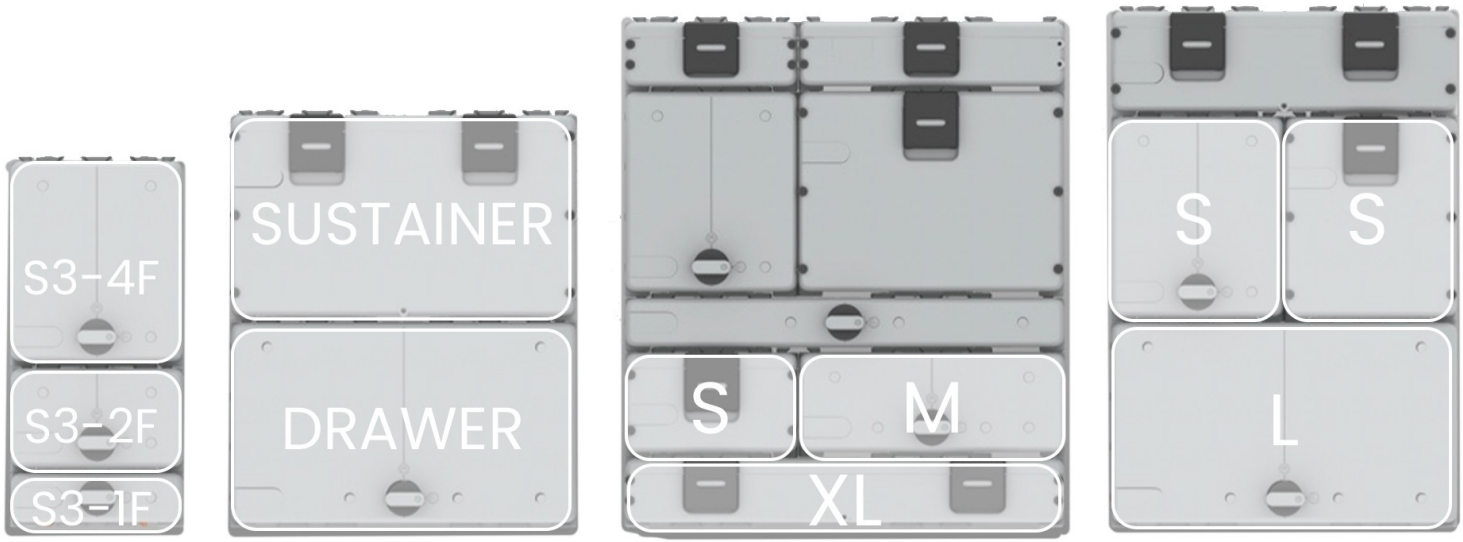
# 160 款模块配置总览





# 堆叠与组合规则



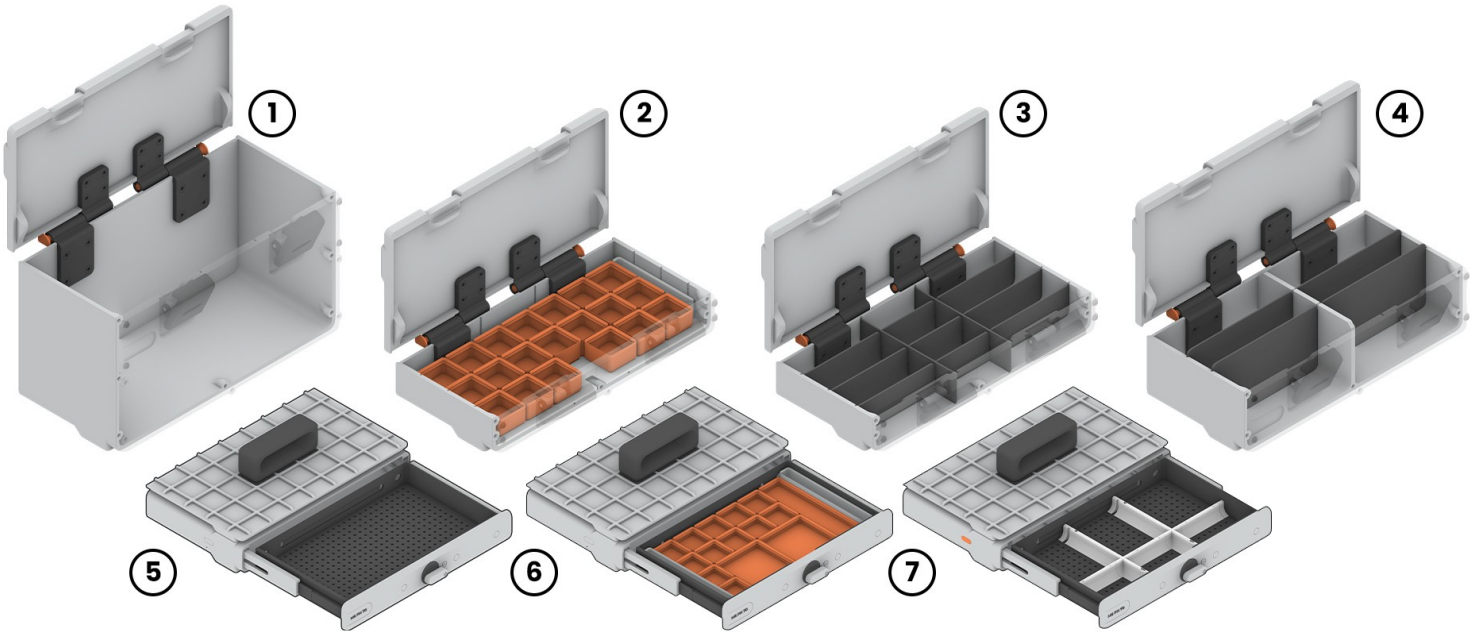



1. Modules with similar depth and width stack on each other.
2. SUstainers and Drawers can stack as well.

1. 相同深宽的模块可垂直叠放；2. 抽屉与收纳盒可混合叠放；3. 宽度 XL = S + M，可组合叠放；4. 宽度 L = S + S，同样稳固互锁。

# 内部隔层与收纳选件

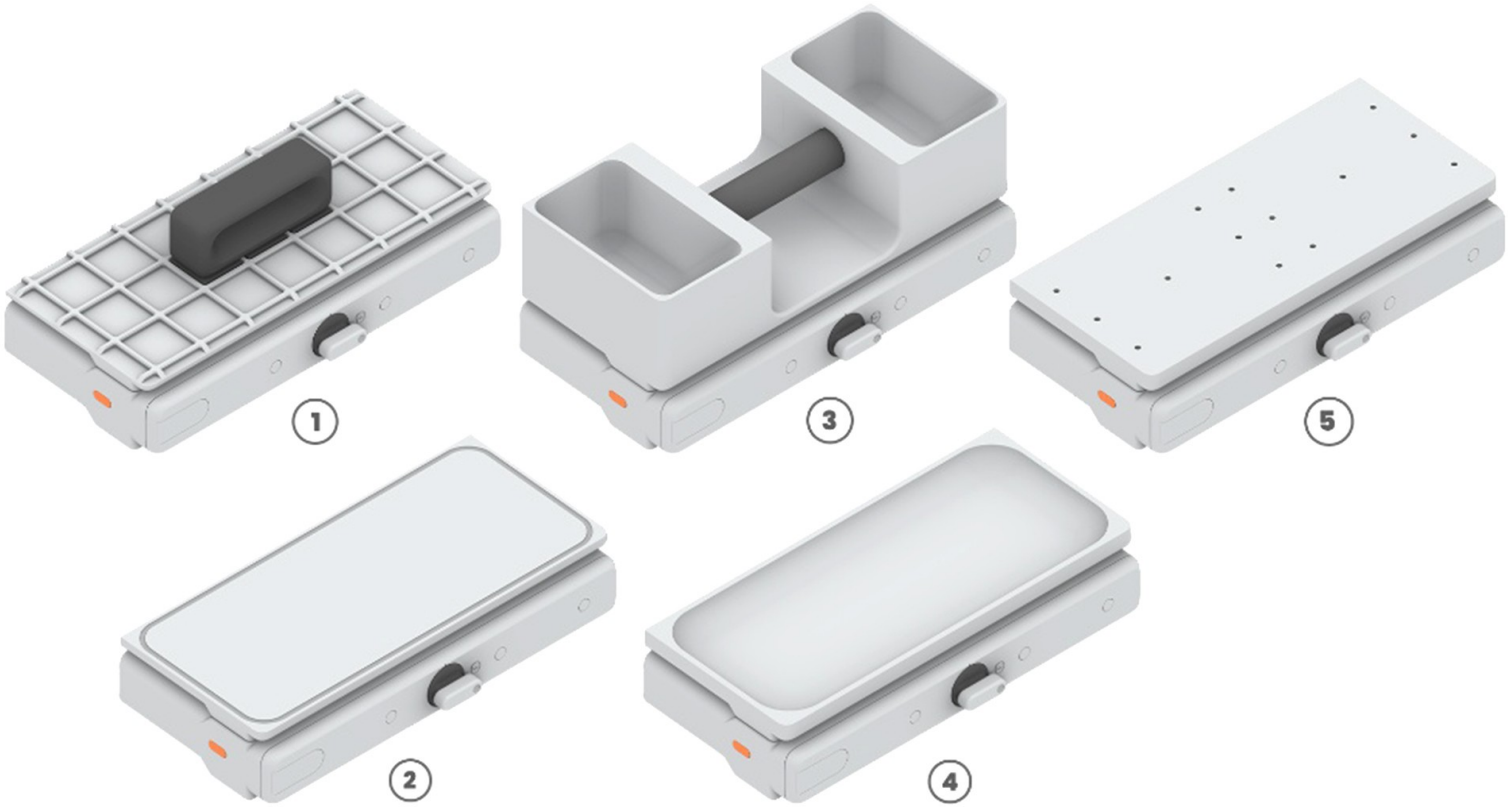





1. 扁平收纳盒　2. Gridfinity 底板　3. 空白抽屉　4. 密封收纳盒　5. 分隔板收纳盒　6. Gridfinity 底座
7. 网格分隔条

# 顶盖类型选件

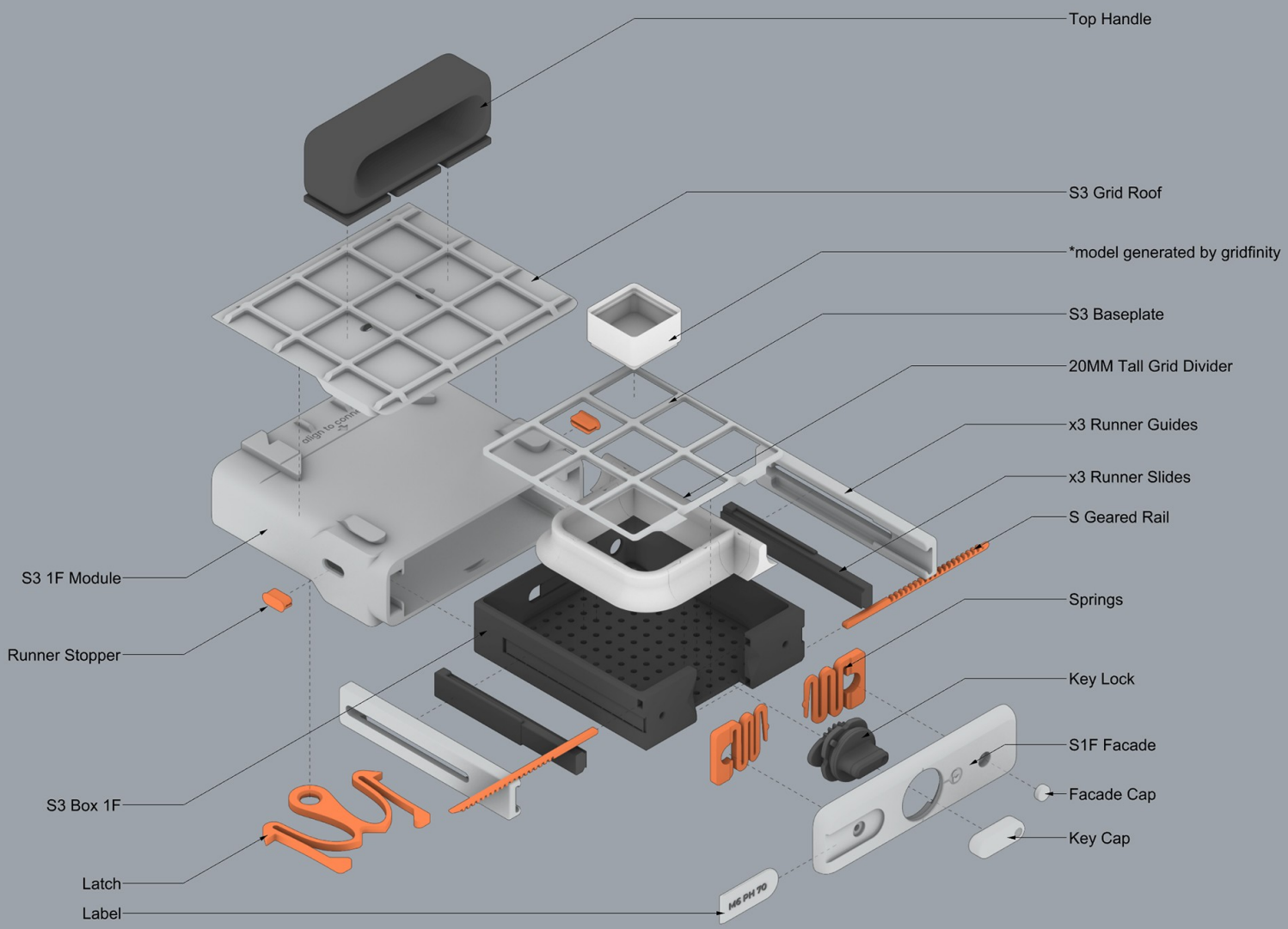





1. 扁平顶板　2. Gridfinity 顶板　3. 承重顶盖嵌件　4. 置物托盘顶板　5. 桌底悬挂安装顶盖

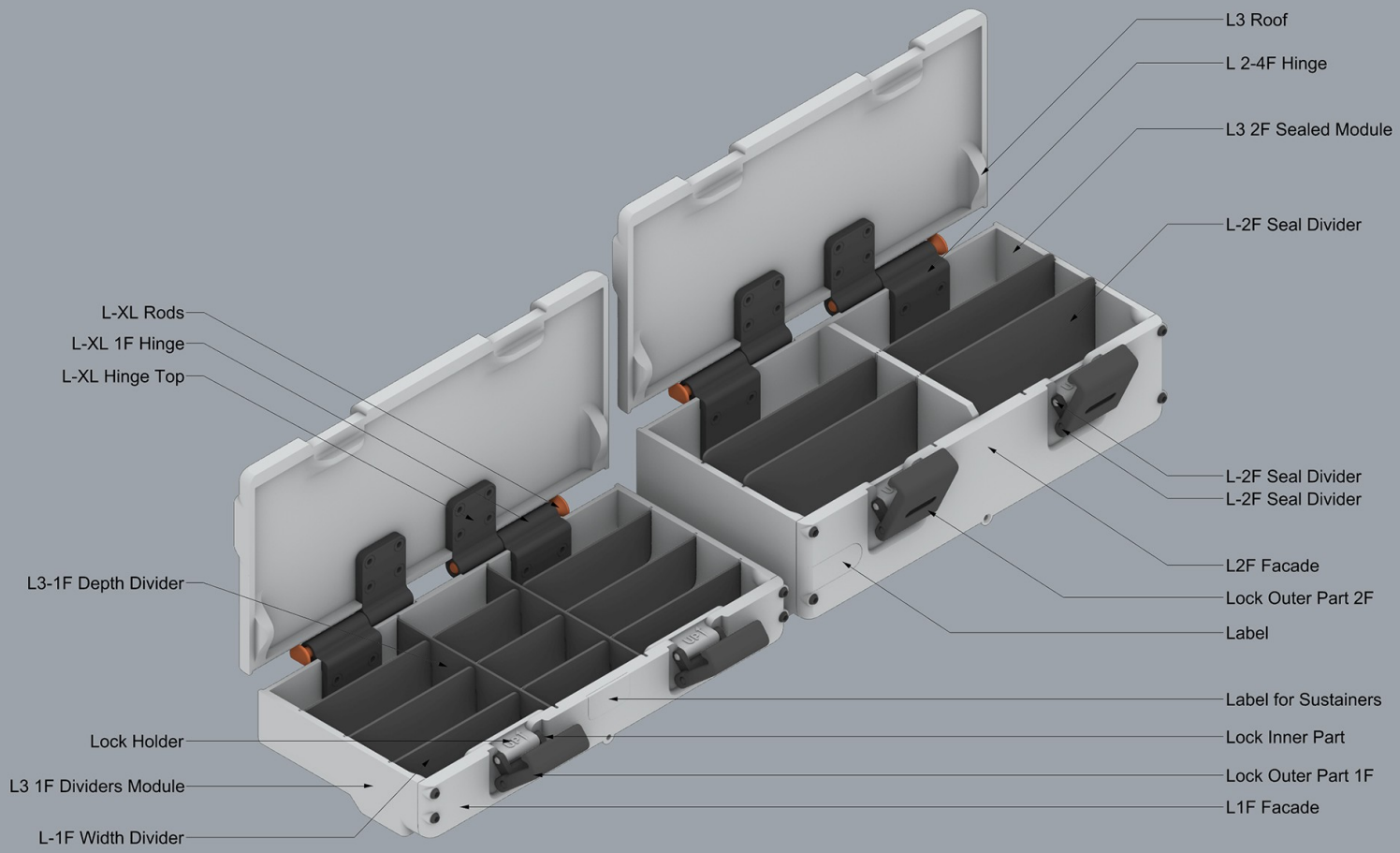
# 抽屉各部件命名对照




Top Handle
S3 Grid Roof
*model generated by gridfinity
S3 Baseplate
20MM Tall Grid Divider
x3 Runner Guides
x3 Runner Slides
S Geared Rail
Springs
Key Lock
S1F Facade
Facade Cap
Key Cap
S3 1F Module
Runner Stopper
S3 Box 1F
Latch
Label
M6 PH 70

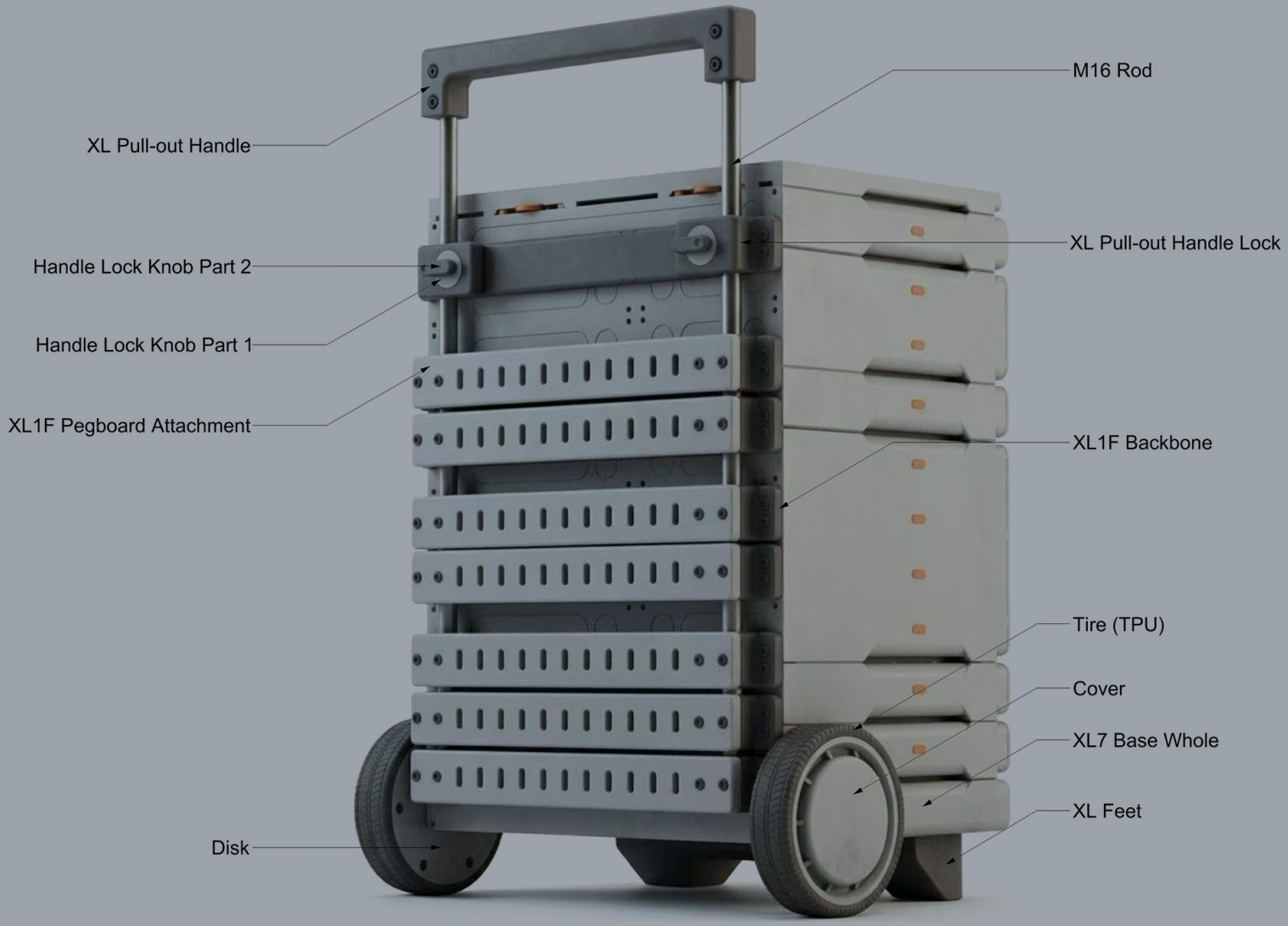
# 收纳盒各部件命名对照




L3 Roof
L 2-4F Hinge
L3 2F Sealed Module
L-2F Seal Divider
L-XL Rods
L-XL 1F Hinge
L-XL Hinge Top
L-2F Seal Divider
L-2F Seal Divider
L2F Facade
Lock Outer Part 2F
Label
L3-1F Depth Divider
Label for Sustainers
Lock Holder
Lock Inner Part
Lock Outer Part 1F
L3 1F Dividers Module
L1F Facade
L-1F Width Divider

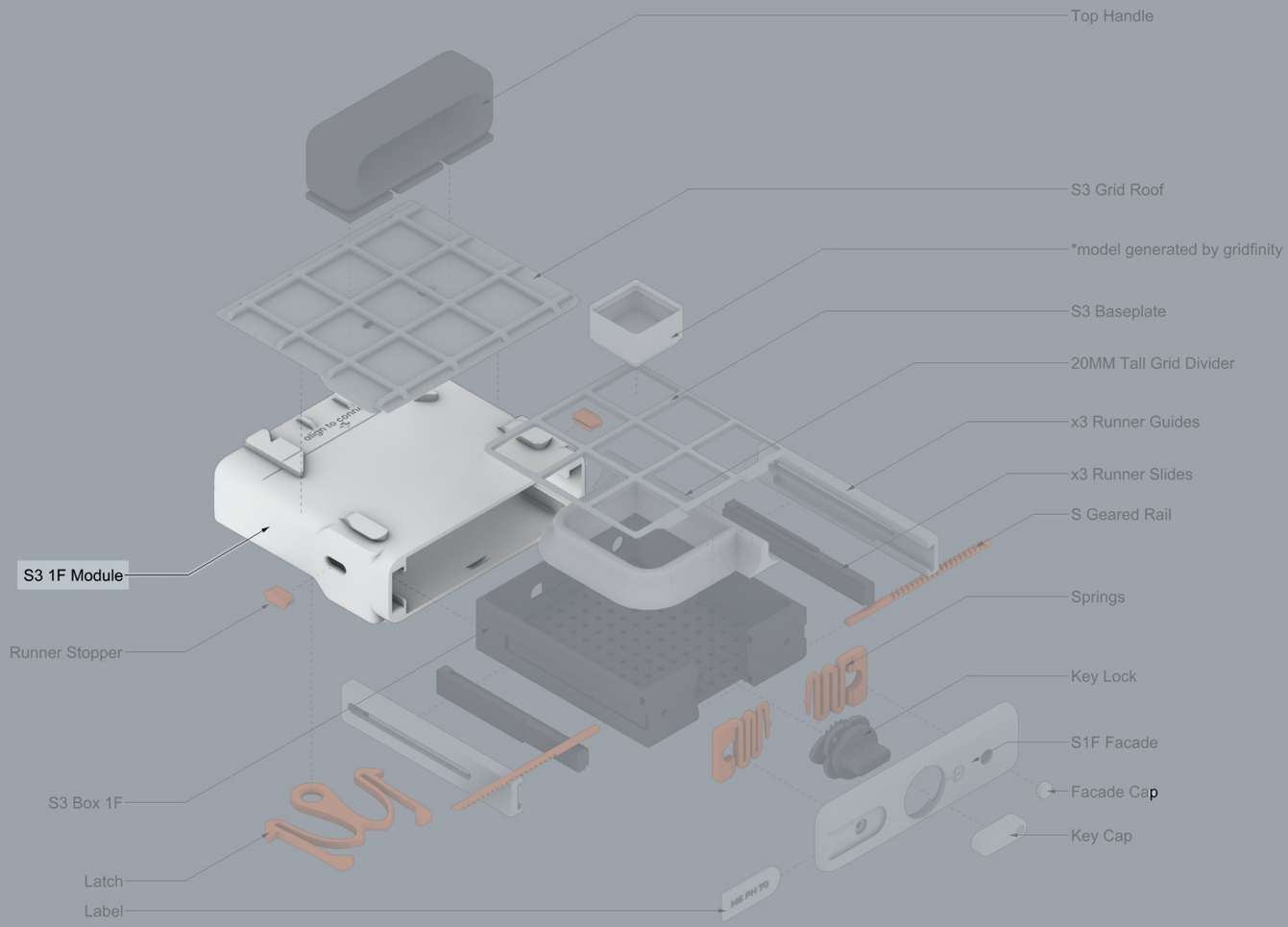
# 扩展改造配件命名对照

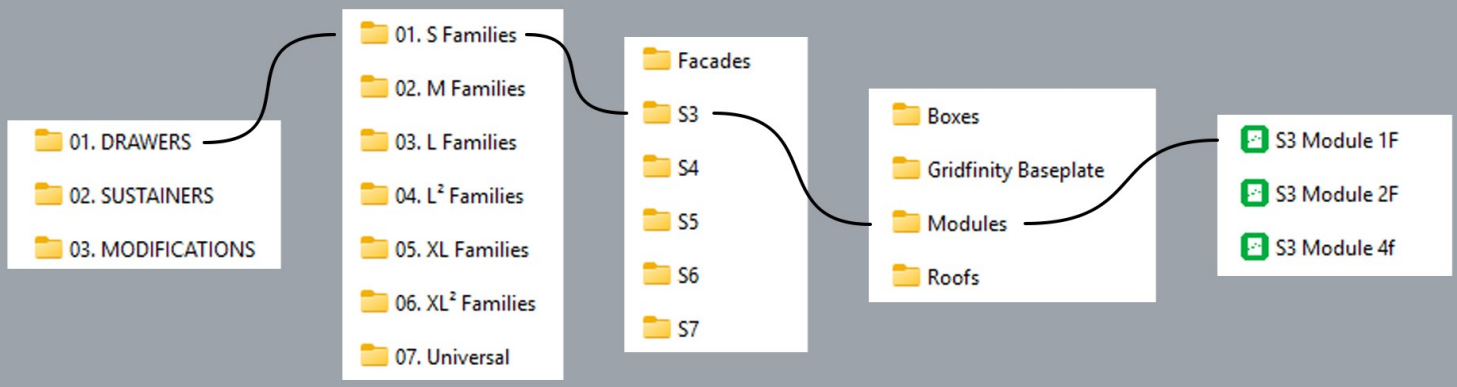



XL Pull-out Handle
M16 Rod
Handle Lock Knob Part 2
XL Pull-out Handle Lock
Handle Lock Knob Part 1
XL1F Pegboard Attachment
XL1F Backbone
Tire (TPU)
Cover
XL7 Base Whole
XL Feet
Disk

# 模型文件目录导览（一）：抽屉与面板

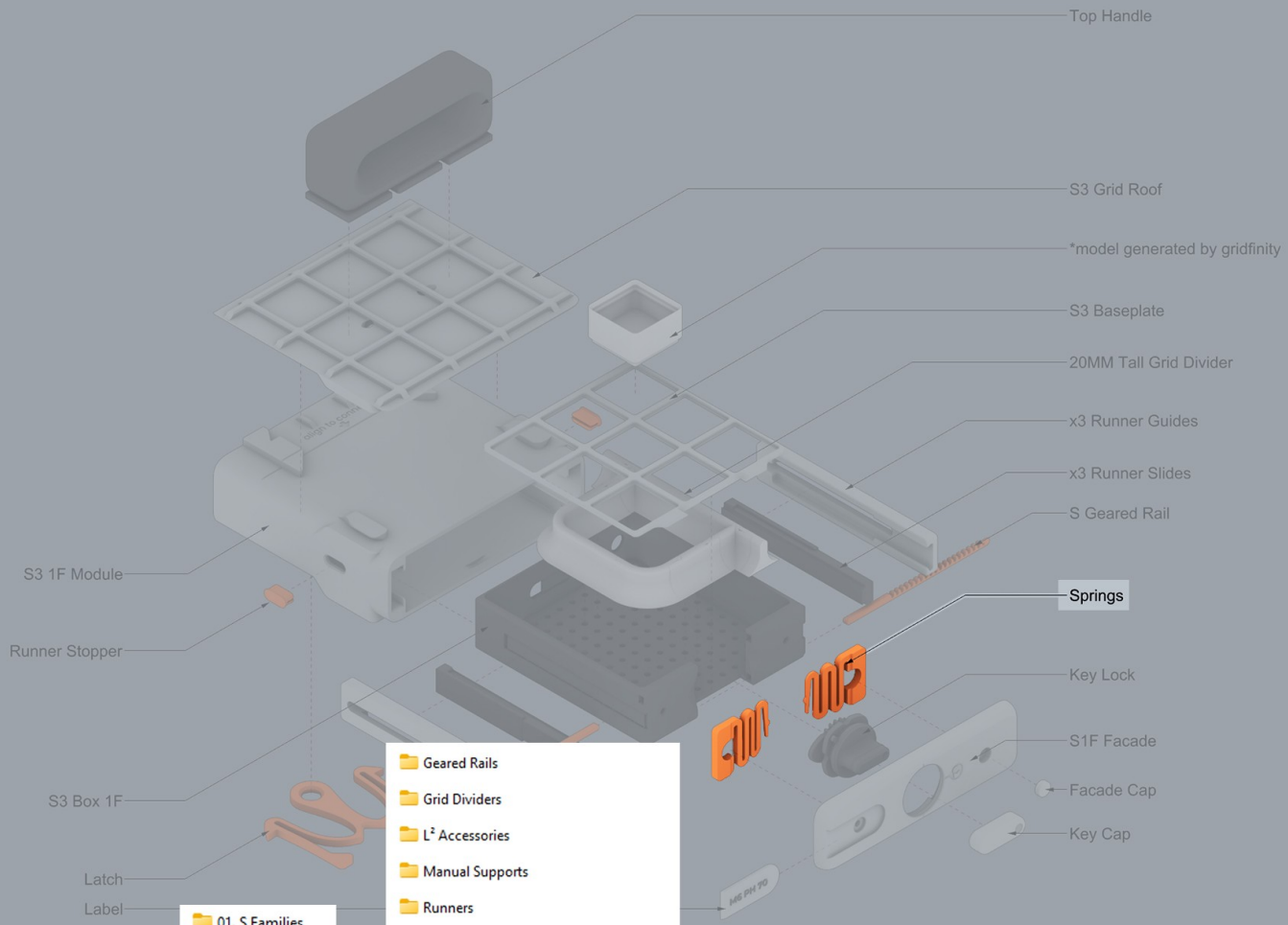




# 模型文件目录导览（二）：滑轨与附件

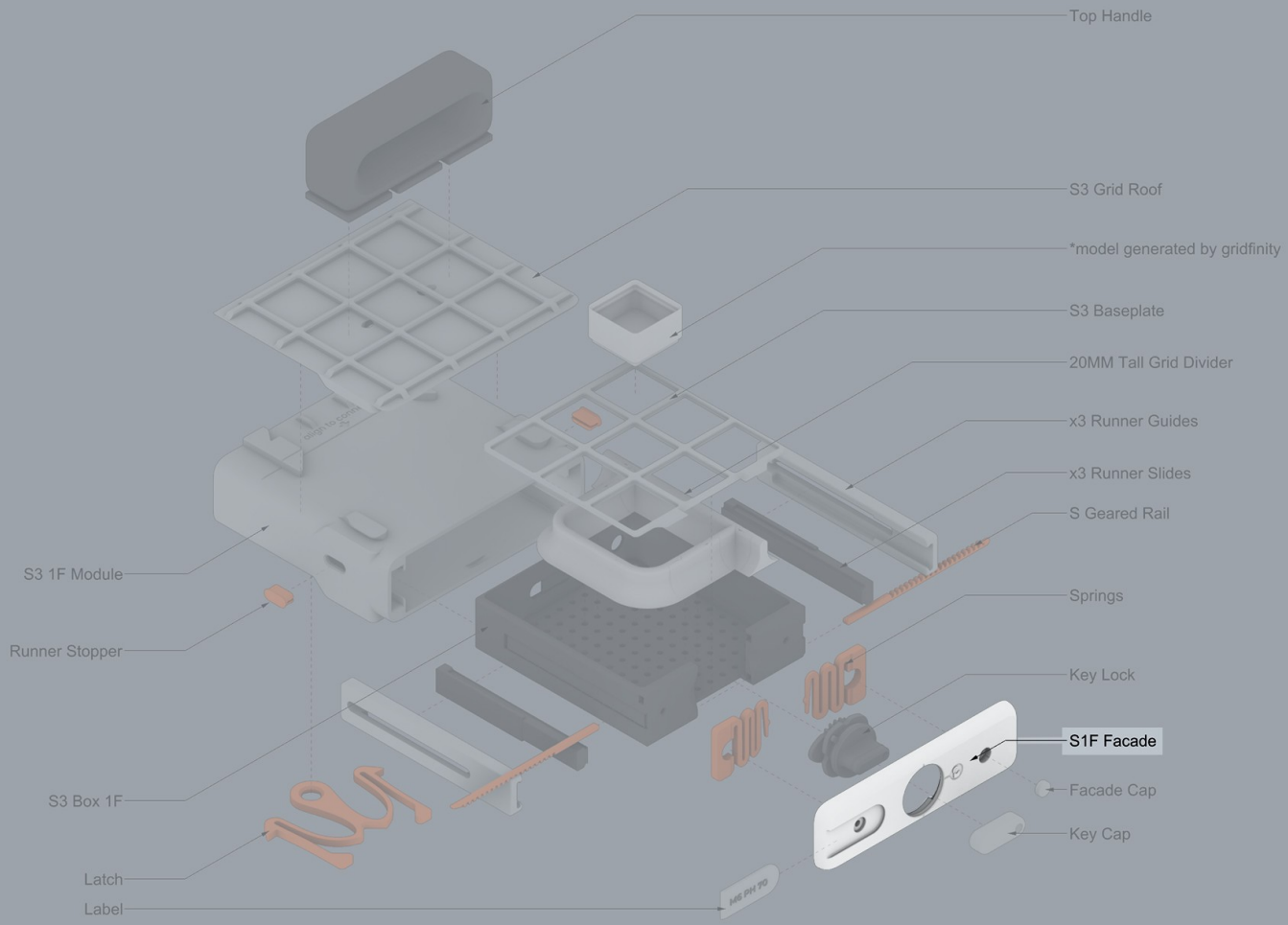


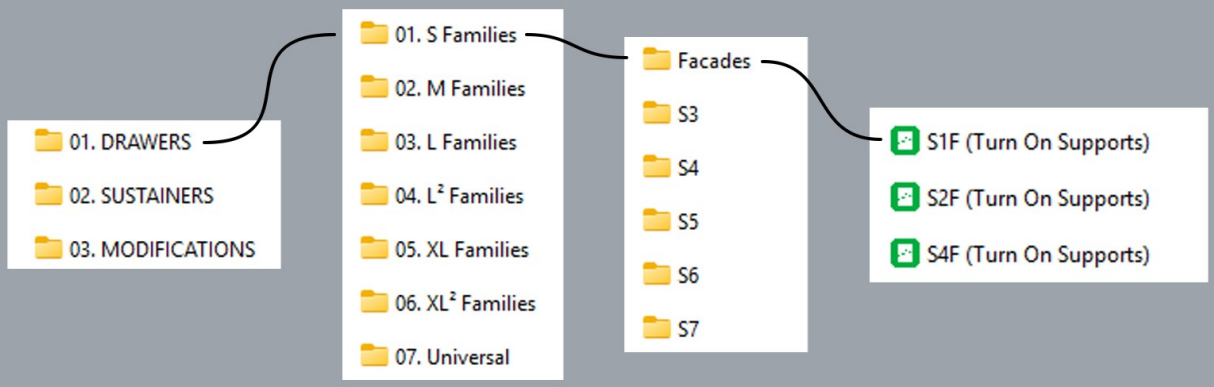



- 01. DRAWERS
- 02. SUSTAINERS
- 03. MODIFICATIONS
  - 01. S Families
  - 02. M Families
  - 03. L Families
  - 04. L² Families
  - 05. XL Families
  - 06. XL² Families
  - 07. Universal
    - Geared Rails
    - Grid Dividers
    - L² Accessories
    - Manual Supports
    - Runners
    - Facade Cap
    - Heawyweight Roof Handle
    - Heawyweight Roof Insert
    - Hidden Key (Turn On Supports)
    - Hidden Key Part 2 (Turn on Supports)
    - Key (Turn On Supports)
    - Key Cap
    - Label
    - Latch
    - Springs
    - Top Handle (Turn On Supports)

# 模型文件目录导览（三）：支撑打印指南

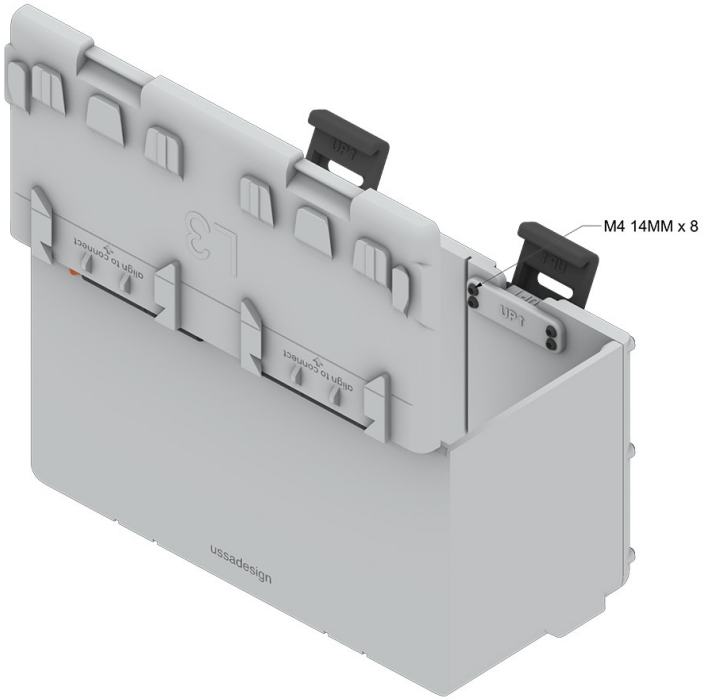


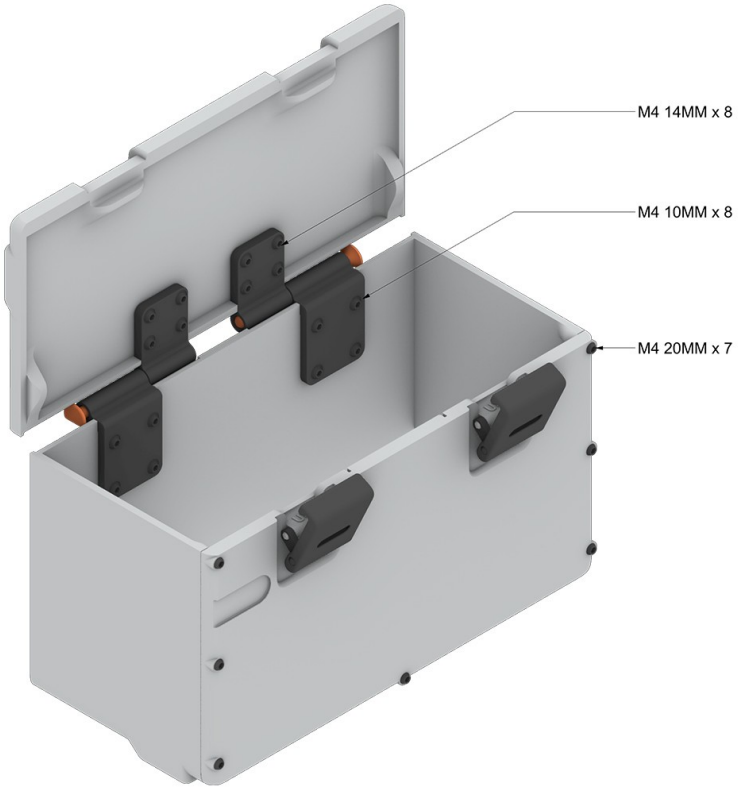


# 收纳盒所需五金配件





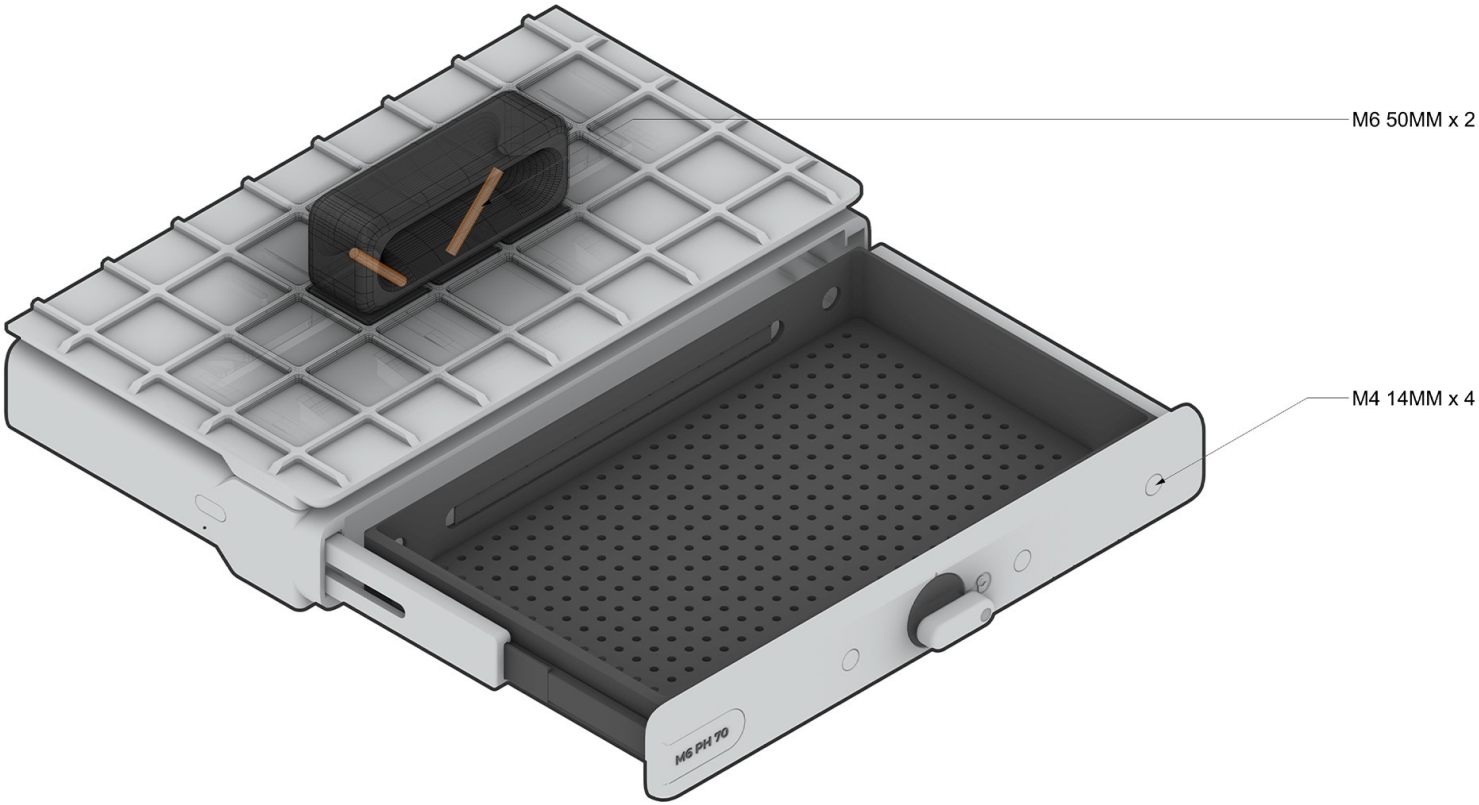




* 每个收纳盒模块所需紧固件：M4×14mm 螺栓 ×16，M4×10mm 螺栓 ×8，M4×20mm 螺栓 ×7

*

# 抽屉所需五金配件



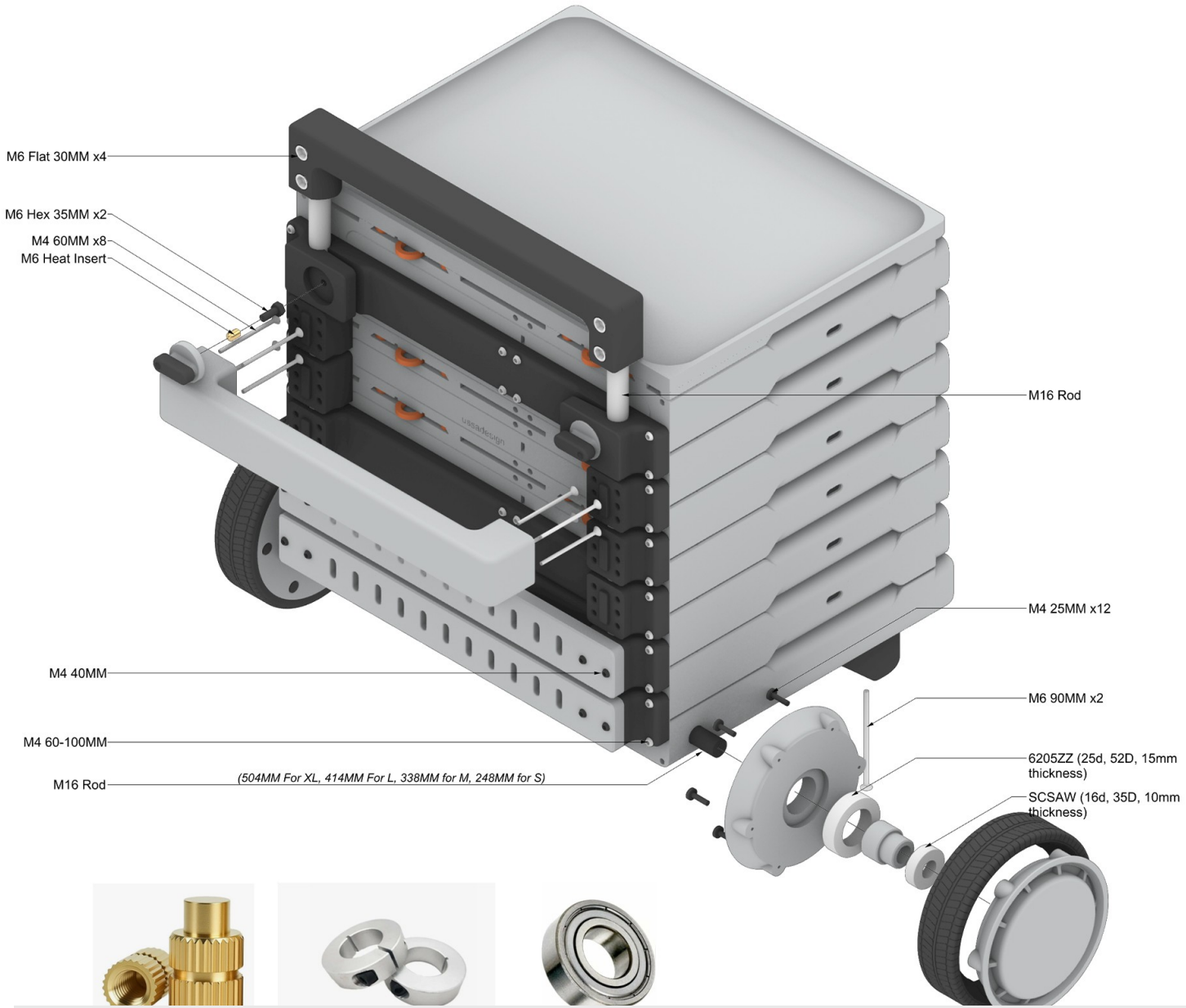



* 抽屉模块所需紧固件：1F 规格需 4 颗 M4×14mm；2F / 4F 规格需 6 颗；每抽屉配 2 颗 M6×50mm 螺栓

*

# 背脊骨架与轮组所需五金配件







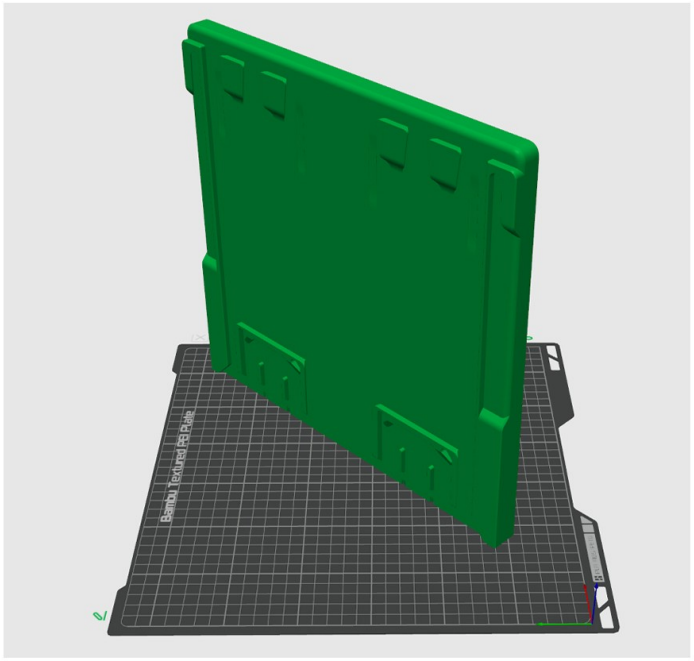


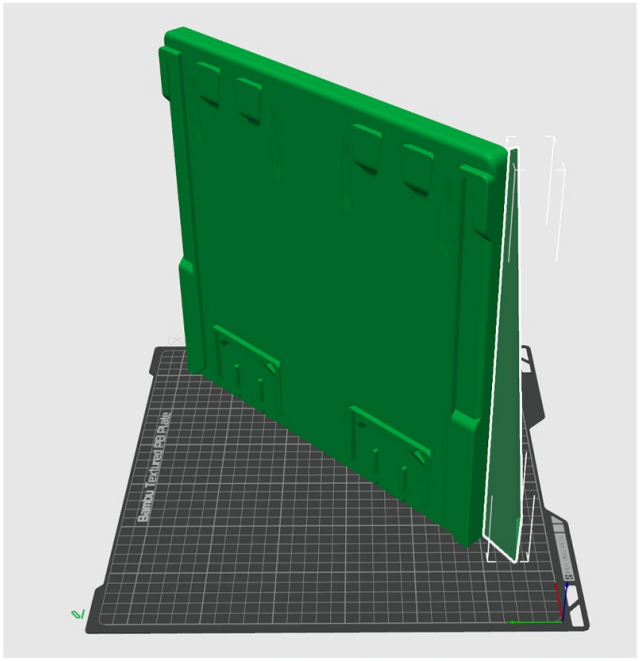


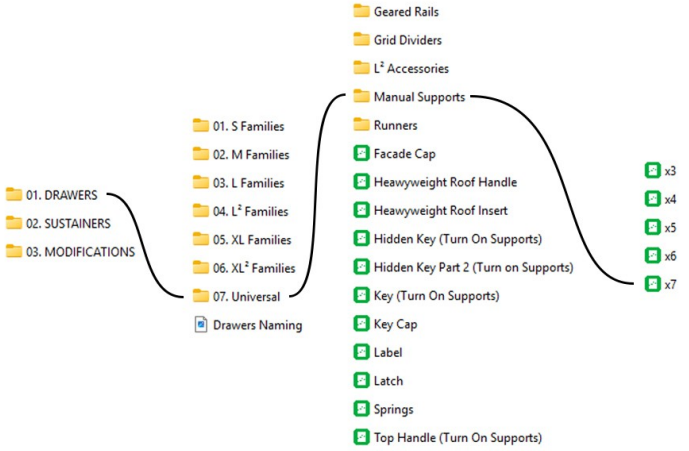

* 背脊与脚轮五金：M6 平头 30mm ×4、M6 六角 35mm ×2、M4 60mm ×8、M6 热熔螺母、M16 螺杆等

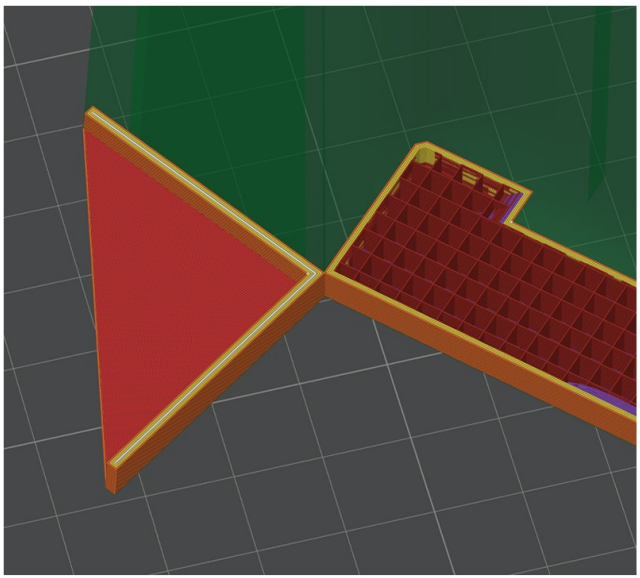
# 手动支撑添加建议





对于较高尺寸的打印件，强烈建议在切片中添加手动支撑，可显著提高成型稳定性并防止晃动变形